

EduPulse: AI-Based Student Success & Dropout Prediction System

Project Overview

EduPulse is a machine learning system that predicts which students are at risk of dropping out or underperforming so educational institutions can intervene early. It combines student academic, attendance, and engagement data to generate a risk score and recommendations.

Objectives

- Predict dropout risk.
- Identify factors contributing to risk.
- Recommend interventions.
- Provide dashboards for students, teachers, and administrators.

Technology Stack

Layer	Technology
Language	Python
ML	Scikit-learn, Pandas, NumPy
Visualization	Matplotlib, Plotly
Backend	Flask/FastAPI
Frontend	HTML, CSS, Bootstrap or React
Database	SQLite/MySQL
Deployment	Streamlit/Render/Railway

Dataset Features

Academic: GPA, internal marks, assignments, quizzes.

Attendance: Attendance %, late arrivals, absences.

Engagement: Class participation, extracurricular activities.

Other: Study hours, internet access, travel distance.

Data Preprocessing

1. Handle missing values.
2. Remove duplicates.
3. Encode categorical variables.
4. Normalize numerical features.
5. Feature engineering (attendance trend, assignment completion ratio, GPA trend).
6. Split data into training and testing sets.

Machine Learning Models

Start with Logistic Regression and Decision Tree. Compare with Random Forest (recommended), XGBoost (advanced), and LightGBM (optional). Use cross-validation and hyperparameter tuning (GridSearchCV/RandomizedSearchCV).

Evaluation Metrics

Accuracy, Precision, Recall, F1 Score, ROC-AUC, Confusion Matrix. Prioritize Recall to identify as many at-risk students as possible.

System Architecture

Student Data → Database → Preprocessing → ML Model → Risk Prediction API → Dashboard → Alerts & Recommendations

Dashboard Modules

Student: Risk score, attendance, performance trends, suggestions.

Teacher: At-risk student list, subject analytics.

Admin: Department-wise trends, dropout statistics, downloadable reports.

Recommendations Engine

Generate personalized actions such as counseling, mentor meetings, revision plans, attendance reminders, and parent notifications based on prediction results.

Project Modules

1. Authentication
2. Student Data Management
3. Data Preprocessing
4. Model Training
5. Prediction Engine
6. Dashboard & Visualization
7. Report Generation
8. Notification Module

Implementation Timeline (8 Weeks)

Week 1: Dataset collection & literature review.

Week 2: Cleaning & preprocessing.

Week 3: Exploratory Data Analysis.

Week 4: Baseline model training.

Week 5: Model comparison & tuning.

Week 6: Backend API.

Week 7: Dashboard development.

Week 8: Testing, documentation, deployment.

Future Enhancements

- Explainable AI using SHAP.
- ERP/LMS integration.
- Mobile application.
- Multilingual support.
- Real-time attendance sync.
- Scholarship and placement prediction.

Resume Value

This project demonstrates end-to-end machine learning, full-stack development, data visualization, deployment, and practical problem solving, making it highly suitable for internships and software/ML roles.